



Python Variables & Data Types

Data with Dylan | Cheat Sheet | Python Fundamentals #1

WHAT IS A VARIABLE?

A variable is a named container that stores a value.

You create one with the = (assignment) operator:

```
my_num = 5
name = "Dylan"
is_active = True
```

💡 Think of it as a labeled box — the name is the label, the value is what's inside.

VARIABLE NAMING RULES

✓ VALID	✗ INVALID
my_score	my score (space)
num1	1num (starts with #)
first_name	first-name (hyphen)
x	class (reserved word)

Convention: use `snake_case` (underscores). Avoid `camelCase`.

TYPE CHECKING & CONVERSION

```
type(5)
# → <class 'int'>
type("hello")
# → <class 'str'>

str(19)
# → "19" (int to string)
int("5")
# → 5 (string to int)
float(3)
# → 3.0 (int to float)
```

⚠ **Must convert** before mixing types in `print()`:

```
print("Age: " + str(age))
# ✓ correct
print("Age: " + age)
# ✗ TypeError
```

⚡ MINI PRACTICE

```
name = "Dylan"
height_inches = 67
Print: "Dylan is 5 feet 7 inches tall"
Hint: use // for feet, % for inches, str() to combine
```

THE 4 MAIN DATA TYPES

TYPE	STORES	EXAMPLE
int	Whole numbers	age = 25
float	Decimal numbers	price = 9.99
str	Text / characters	name = "Dylan"
bool	True or False only	logged_in = True

KEY SYNTAX & OPERATIONS

Integers & Floats:

```
5 + 5
# → 10
5 / 2
# → 2.5 (always a float)
5 // 2
# → 2 (integer division)
5 % 2
# → 1 (remainder)
```

Strings:

```
first = "Dylan"
last = "Song"
print(first + " " + last)
# → Dylan Song
```

Booleans:

```
is_raining = True
logged_in = False
```

💡 Booleans are most useful inside `if` statements (coming later!)

COMMON BEGINNER MISTAKES

✗ **Space in a variable name**

```
my score = 100
✓ my_score = 100
```

✗ **Mixing quote types in a string**

```
name = 'Dylan' → use consistent quotes
✓ name = "Dylan"
```

✗ **Concatenating a non-string**

```
print("Age: " + age)
✓ print("Age: " + str(age))
```

✗ **Confusing = vs ==**

```
= assigns a value
== compares values
```